

# S&P Global AI Engineering - Recommended Tools & Platforms

The recommended core toolkit for the AI Engineering group at S&P Global is Python 3.6, including the various modules and libraries that make up the “Python Data Science Ecosystem” (sometimes called the “Python Scientific Computing Stack”):

- [NumPy](#) & [SciPy](#) for mathematical functions
- [Pandas](#) for data preparation and basic analysis
- [matplotlib](#) & [Bokeh](#) for data visualization
- [scikit-learn](#) for machine learning
- [PyTorch](#) for deep learning
- [spaCy](#) & [NLTK](#) for text analysis
- [PySpark](#) (Apache Spark) and [Dask](#) for big data analysis
- etc.

The recommended distribution of Python to use for learning and experimentation is the [Anaconda distribution](#) for Python 3.6. Anaconda is free, widely-used, and comes with many of the core Python data science libraries pre-installed. Information about installing Anaconda can be found in the “Python Setup & Skills” section of the [SPGMI Data Science training curriculum](#).

The [Jupyter Notebook](#) (formerly known as iPython Notebook) is a recommended development application for performing data science work. The Notebook offers a web-based, interactive interface for writing and executing code in Python (or a variety of other languages), as well as mixing functional code in with rich text, images, and other web content. A quick demo of some Jupyter Notebook functionality is also listed in the “Python Setup & Skills” section of the [SPGMI Data Science training curriculum](#).

## Why Python?

Python, along with its related data science ecosystem, has a number of attractive features as a core data science toolkit:

- It’s free and open source
- It’s widely used and actively developed for data science work in industry
- It’s a general purpose programming language — you can do more with a single tool and reduce the need to switch between multiple tools
- It’s considered easier to learn for people who are already familiar with another programming language (C#, Java, etc.)
- It’s considered easier to integrate Python code with existing production systems

## Domino Data Lab

S&P Global has licensed [Domino Data Lab](#) as our data science development & management platform for the AI Engineering group. Domino takes the functionality of the Python data science ecosystem and wraps it with enterprise-oriented features, such as:

- Automated version control optimized for data science workflows (code, data, models)
- Simple creation of REST APIs to deploy data science models and integrate them with other products and systems
- Elastic scalability of cloud computing resources to support data science workflows of various levels of computational intensity
- Collaboration features to enable multiple data scientists to contribute to a shared project base

Data science work at SPGMI that's for training or experimental purposes may be conducted on local workstations or development servers using Anaconda. Projects that are eventually intended to be deployed in a "production" context should migrate into Domino at an appropriate point in their development cycle.

For more information about Domino, please consult their [website](#), [support page](#), or their [blog](#).